



System calls made by a command

The `strace` utility is used to display system calls made by a command, so you can see exactly what the command is doing until it exits. This helps in debugging of many applications. The output is printed on the terminal, which can be customised based on the options available in the manual pages.

For example, with:

```
$ strace who
```

...the output will be

```
execve("/usr/bin/who", ["who"], [/ 38 vars /]) = 0
brk(0) = 0xb691000
access("/etc/ld.so.nohwcap", F_OK) = -1 ENOENT (No
such file or directory)
mmap2(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_
ANONYMOUS, -1, 0) = 0xb7768000
access("/etc/ld.so.preload", R_OK) = -1 ENOENT (No
such file or directory)
open("/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat64(3, {st_mode=S_IFREG|0644, st_size=61896, ...}) = 0
mmap2(NULL, 61896, PROT_READ, MAP_PRIVATE, 3, 0) =
0xb7758000
close(3) = 0
access("/etc/ld.so.nohwcap", F_OK) = -1 ENOENT (No
such file or directory)
```

If you want only the count of system calls, run the following command:

```
$ starce -C who
```

This will display the count of the system calls and errors.

—Suresh S Patil,
suresh.patil106@gmail.com

Tips on using VI editor in a UNIX/Linux environment

In UNIX/Linux environments, VI editor is one of the most

powerful and commonly used. The following tips can make an admin or developer's job easier.

Go to VI while using more

Often, we come across situations in which we go through the files using the `more` command, when we suddenly want to modify the contents. The following tip will be very useful in such situations.

While paging through a file in `more`, press `v` to start editing the file in VI at the current position in the file. After pressing `v`, VI editor will be invoked from `more`, however when you come out from VI by using `:wq`, you will come back to paging through a file in `more`. You will be able to see the changes that you have made.

Jump to a line number when editing a file with VI:

Most of the time, when we compile files in the UNIX/Linux environment, the compiler will throw an error on the standard output or will log it in the associated log files. We often come across errors mentioning the line number with the error, as shown below.

syntax error on line 2070.

Paranthesis missing error on line 212.

The following tip will be very useful in such situations. Run the command shown below:

```
vi filename +linenumber
```

This will open up the file with the cursor automatically moved down to the specified line. You can jump straight to the problem without fumbling around. And you can even go to any other line in VI editor by just typing the following command while using VI.

```
:linenumber
```

Search and replace when editing a file with VI

The 'search and replace' option in VI is the most commonly used. Let us look at some simple options that can be used for searching a string forwards and